Lesson 11.3 - Hypothesis Testing with Two Proportions

> **Video:** Hypothesis Testing With Two Proportions by The Organic Chemistry Tutor **Chapter:** 11 - Hypothesis Testing - 2 Samples

The big picture: comparing two rates

Lessons 11.1 and 11.2 compared **averages** (scores, weights). Many questions are about **yes/no** data instead: defective or not, passed or failed, clicked or didn't. Each group is then summarized by a **proportion** (a rate), and we ask whether the two rates really differ.

1. The question. Batch 1 has 4% defective laptops and batch 2 has 6%. Is batch 2 really worse, or could a 2-point gap happen by chance with samples this size?

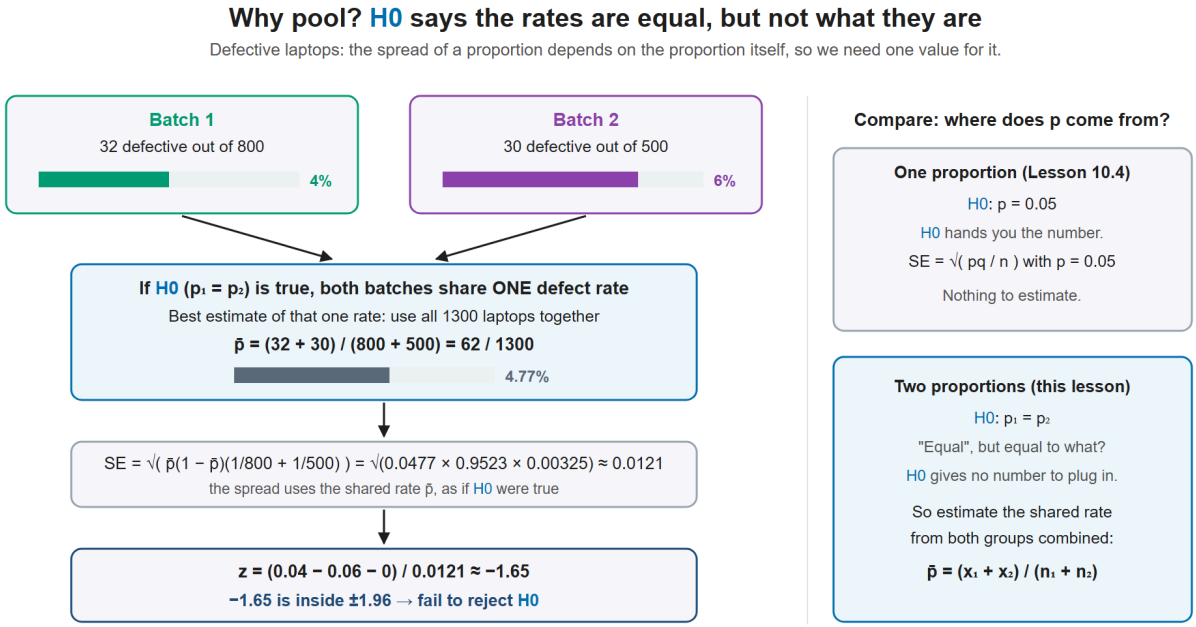
2. H_0 says "no difference". $H_0 : p_1 = p_2$, which is the same as $p_1 - p_2 = 0$. The recipe is the same as always:

$$z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{SE}$$

3. Why pool? H_0 says "equal", but not equal to what. For proportions, the spread depends on the proportion itself: $SE = \sqrt{p(1-p)/n}$. A rate near 50% wobbles more than a rate near 4%. So we need a value of p to compute the SE.

- **One proportion (Lesson 10.4):** $H_0 : p = 0.05$ hands us the number. Plug it in.
- **Two proportions (this lesson):** $H_0 : p_1 = p_2$ gives no number. But if H_0 is true, both groups share **one** rate, and the best estimate of that rate uses **all** the data together:

$$\bar{p} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{\text{total successes}}{\text{total sample size}}$$



If H_0 is true, the two batches are really one population, so we estimate its single defect rate from all 1300 laptops. That pooled rate goes into the standard error.

4. Why no pooling for means (Lesson 11.1)? A mean and its spread are separate numbers. $H_0 : \mu_1 = \mu_2$ says nothing about s_1 and s_2, so each group keeps its own. For proportions, the spread is tied to p, so assuming equal p's also means assuming equal spreads.

5. Why z, not t? There is no separate s to estimate, because the spread comes straight from $\bar{p}$. With large samples the sample proportions are approximately normal, so we use z. A common check is at least about 10 successes and 10 failures in each group. Here there are 32 and 30 defectives, so that's fine.

Formulas

Quantity	Formula
Sample proportions	$\hat{p}_1 = \dfrac{x_1}{n_1},\ \hat{p}_2 = \dfrac{x_2}{n_2}$
Pooled proportion	$\bar{p} = \dfrac{x_1 + x_2}{n_1 + n_2}$
Test statistic	$z = \dfrac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\bar{p}\,(1-\bar{p})\left(\dfrac{1}{n_1} + \dfrac{1}{n_2}\right)}}$

Under $H_0 : p_1 = p_2$, the term $p_1 - p_2$ is **0**. The proportions are pooled because H_0 assumes both groups share one common proportion. Large samples → use the normal (z) distribution.

Example - defective laptops

Two batches of laptops were tested: 32 of 800 were defective in group 1 and 30 of 500 in group 2. Is the difference significant at $\alpha = 0.05$?

Hypotheses (the test is run because someone suspects a difference):

$$H_0 : p_1 = p_2 \qquad H_a : p_1 \neq p_2$$

Sample proportions:

$$\hat{p}_1 = \frac{x_1}{n_1} = \frac{32}{800} = 0.04, \qquad \hat{p}_2 = \frac{x_2}{n_2} = \frac{30}{500} = 0.06$$

Critical values (two-tailed, $\alpha/2 = 0.025$ per tail, left area 0.975): $z = \pm 1.96$

Pooled proportion:

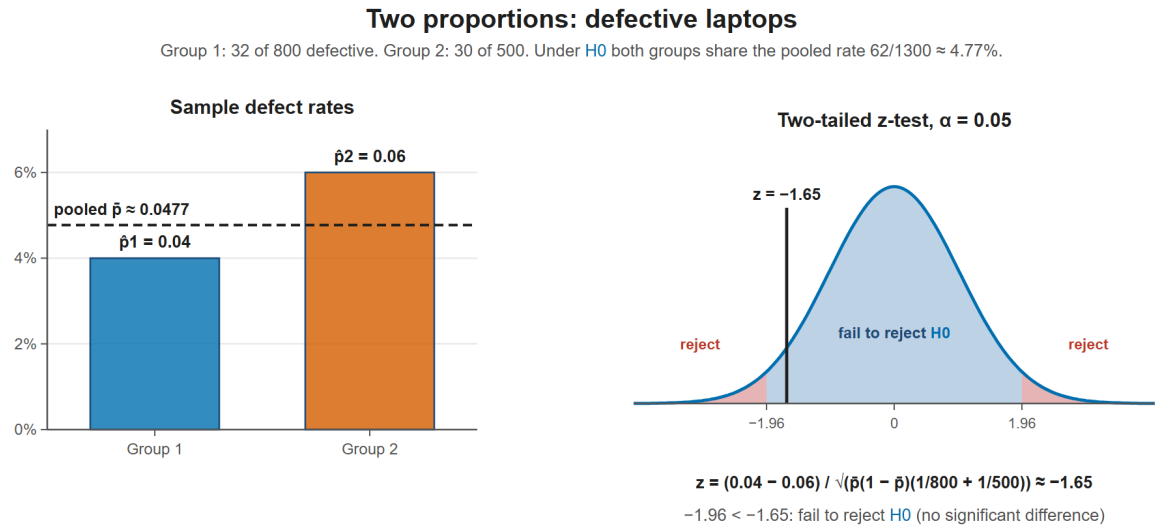
$$\bar{p} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{32 + 30}{800 + 500} = \frac{62}{1300} \approx 0.04769$$

Test statistic:

$$z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{\sqrt{\bar{p}(1-\bar{p})\left(\dfrac{1}{n_1} + \dfrac{1}{n_2}\right)}} = \frac{0.04 - 0.06 - 0}{\sqrt{0.04769(1 - 0.04769)\left(\dfrac{1}{800} + \dfrac{1}{500}\right)}} = \frac{-0.02}{0.012149} \approx -1.65$$

$-1.96 < -1.65 < 0$, which is in the fail-to-reject region.

Fail to reject H_0. At the 5% level the difference in defect rates (4% vs 6%) is **not statistically significant**. This doesn't prove the rates are equal; the data just don't show a clear difference.



The two sample rates differ by 2 percentage points, but with these sample sizes that gives only z ≈ −1.65, which is inside ±1.96. The difference is not statistically significant.

Key takeaways

- We pool because H_0 says the rates are equal without saying what they equal, and the SE of a proportion needs a value of p.
- H_0 : $p_1 = p_2$. Use the **pooled** proportion $\bar{p} = (x_1 + x_2)/(n_1 + n_2)$ in the standard error.
- $z = (\hat{p}_1 - \hat{p}_2)/\sqrt{\bar{p}(1-\bar{p})(1/n_1 + 1/n_2)}$.
- Compare with the critical value(s) or use a p-value. Fail to reject means "no significant difference found", not "proven equal".